Minnesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1162. URL https://aclanthology.org/N19-1162.

Freda Shi, Daniel Fried, Marjan Ghazvininejad, Luke Zettlemoyer, and Sida I Wang. Natural language to code translation with execution. *arXiv preprint arXiv:2204.11454*, 2022a.

Freda Shi, Kevin Gimpel, and Karen Livescu. Substructure distribution projection for zero-shot cross-lingual dependency parsing. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 6547–6563, Dublin, Ireland, May 2022b. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.452. URL https://aclanthology.org/2022.acl-long.452.

Haoyue Shi, Luke Zettlemoyer, and Sida I. Wang. Bilingual lexicon induction via unsupervised bitext construction and word alignment. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 813–826, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.67. URL https://aclanthology.org/2021.acl-long.67.

Taylor Shin, Yasaman Razeghi, Robert L. Logan IV, Eric Wallace, and Sameer Singh. AutoPrompt: Eliciting Knowledge from Language Models with Automatically Generated Prompts. *EMNLP*, 2020. doi: 10.18653/v1/2020.emnlp-main.346. URL https://aclanthology.org/2020.emnlp-main.346.

Georgios Spithourakis and Sebastian Riedel. Numeracy for language models: Evaluating and improving their ability to predict numbers. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 2104–2115, Melbourne, Australia, July 2018. Association for Computational Linguistics. doi: 10.18653/v1/P18-1196. URL https://aclanthology.org/P18-1196.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, and Denny Zhou. Rationale-augmented ensembles in language models. *arXiv preprint arXiv:2207.00747*, 2022.

Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, et al. Emergent abilities of large language models. *Transactions on Machine Learing Research (TMLR)*, 2022a. URL https://arxiv.org/abs/2206.07682.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain of thought prompting elicits reasoning in large language models. *Conference on Neural Information Processing Systems (NeurIPS)*, 2022b. URL https://arxiv.org/abs/2201.11903.

Genta Indra Winata, Andrea Madotto, Zhaojiang Lin, Rosanne Liu, Jason Yosinski, and Pascale Fung. Language Models are Few-shot Multilingual Learners. In *Proceedings ofthe 1st Workshop on Multilingual Representation Learning*, 2021. URL http://arxiv.org/abs/2109.07684.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. mT5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 483–498, Online, June 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.naacl-main.41. URL https://aclanthology.org/2021.naacl-main.41.

Mengjie Zhao and Hinrich Schütze. Discrete and Soft Prompting for Multilingual Models. In *Proceedings of EMNLP 2021*, pp. 8547–8555, 2021. URL http://arxiv.org/abs/2109.03630.

	en	de	fr	es	ru	zh	ja	th	te	bn	sw
# Exemplars	6	6	6	6	1	5	4	1	1	1	6

Table 7: Number of few-shot exemplars for GPT-3 experiments in Table 3.

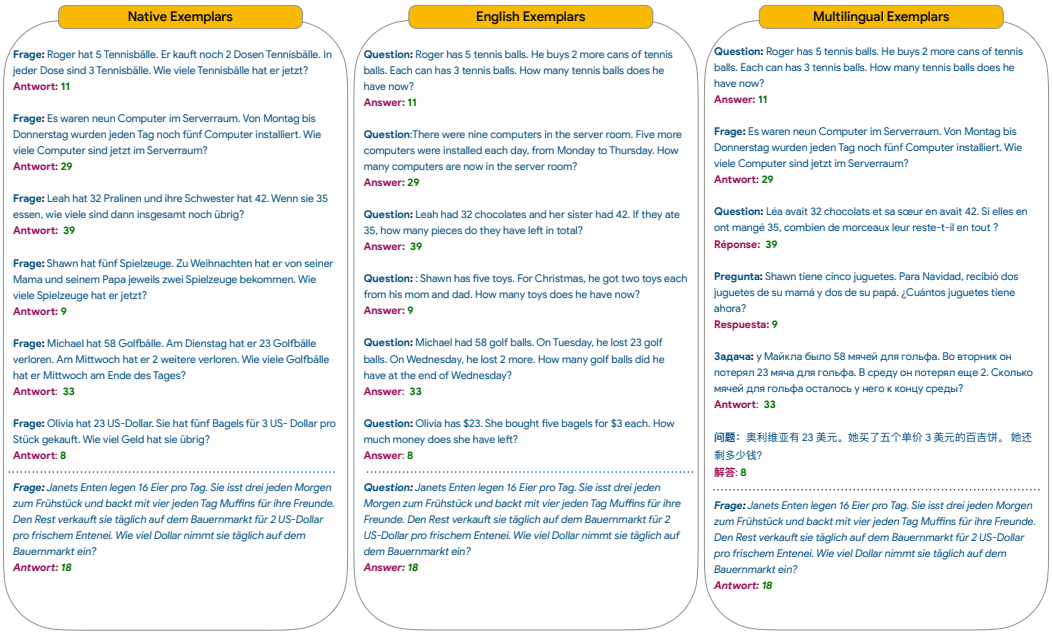

Figure 6: Prompt template in the direct answer prediction setting (DIRECT), solving a problem in German. Above dotted lines: few-shot exemplars; below dotted lines: the question of interest and the expected answer. The dotted lines are not included in our experiments.

A DETAILS OF MGSM EXPERIMENTS

In this section, we present details of our experiments on MGSM, including the number of exemplars used for GPT-3 (§A.1) and the detailed prompts in each setting summarized in Table 2 (§A.2).

A.1 NUMBER OF EXEMPLARS FOR EACH LANGUAGE

Given the unbalanced representation of languages in the training corpora, the byte-pair encoding (BPE; Gage, 1994) algorithm tokenizes sentences in underrepresented languages, especially those in a different alphabet from English, into more tokens. Given that the GPT-3 API supports a maximum number of 2048 tokens as its input, it does not support 6-shot prompting in some languages, including Russian, Chinese, Japanese, Thai, Telugu and Bengali; therefore, we use the maximum possible number of exemplars (Table 7) instead for GPT-3, while using 6-shot for all languages in PaLM experiments.

A.2 MGSM PROMPTS IN EACH SETTING

We present the prompts used in our MGSM experiments in Figures 6 to 8, where the TRANSLATE-EN experiments can be viewed as a English one with EN-COT and ENGLISH-EXEMPLARS.

B DETAILED MGSM PERFORMANCE

We report the detailed numbers in our analysis (Figures 4 and 5) in Table 8.

Native Exemplars

Frage: Roger hat 5 Tennisbälle. Er kauft noch 2 Dosen Tennisbälle. In jeder Dose sind 3 Tennisbälle. Wie viele Tennisbälle hat er jetzt?
Step-by-Step Answer: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is **11**.

Frage: Es waren neun Computer im Serverraum. Von Montag bis Donnerstag wurden jeden Tag noch fünf Computer installiert. Wie viele Computer sind jetzt im Serverraum?
Step-by-Step Answer: There are 4 days from Monday to Thursday. 5 computers were added each day. That means in total 4 * 5 = 20 computers were added. There were 9 computers in the beginning, so now there are 9 + 20 = 29 computers. The answer is **29**.

Frage: Leah hat 32 Pralinen und ihre Schwester hat 42. Wenn sie 35 essen, wie viele sind dann insgesamt noch übrig?
Step-by-Step Answer: Leah had 32 chocolates and Leah's sister had 42. That means there were originally 32 + 42 = 74 chocolates. 35 have been eaten. So in total they still have 74 - 35 = 39 chocolates. The answer is **39**.

Frage: Shawn hat fünf Spielzeuge. Zu Weihnachten hat er von seiner Mama und seinem Papa jeweils zwei Spielzeuge bekommen. Wie viele Spielzeuge hat er jetzt?
Step-by-Step Answer: He has 5 toys. He got 2 from mom, so after that he has 5 + 2 = 7 toys. Then he got 2 more from dad, so in total he has 7 + 2 = 9 toys. The answer is **9**.

Frage: Michael hat 58 Golfbälle. Am Dienstag hat er 23 Golfbälle verloren. Am Mittwoch hat er 2 weitere verloren. Wie viele Golfbälle hat er Mittwoch am Ende des Tages?
Step-by-Step Answer: Michael started with 58 golf balls and lost 23, so he has 58 - 23 = 35. After he lost 2 more, he has 35 - 2 = 33 balls now. The answer is **33**.

Frage: Olivia hat 23 US-Dollar. Sie hat fünf Bagels für 3 US- Dollar pro Stück gekauft. Wie viel Geld hat sie übrig?
Step-by-Step Answer: 5 bagels for $3 each should cost 5 * 3 = 15 dollars. Olivia had $23 in the beginning, so now she has 23 - 15 = 8 dollars left. The answer is **8**.

Frage: *Janets Enten legen 16 Eier pro Tag. Sie isst drei jeden Morgen zum Frühstück und backt mit vier jeden Tag Muffins für ihre Freunde. Den Rest verkauft sie täglich auf dem Bauernmarkt für 2 US-Dollar pro frischem Entenei. Wie viel Dollar nimmt sie täglich auf dem Bauernmarkt ein?*
Step-by-Step Answer: *[...] The answer is* ***18***.

English Exemplars

Question: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
Step-by-Step Answer: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is **11**.

Question: There were nine computers in the server room. Five more computers were installed each day, from Monday to Thursday. How many computers are now in the server room?
Step-by-Step Answer: There are 4 days from Monday to Thursday. 5 computers were added each day. That means in total 4 * 5 = 20 computers were added. There were 9 computers in the beginning, so now there are 9 + 20 = 29 computers. The answer is **29**.

Question: Leah had 32 chocolates and her sister had 42. If they ate 35, how many pieces do they have left in total?
Step-by-Step Answer: Leah had 32 chocolates and Leah's sister had 42. That means there were originally 32 + 42 = 74 chocolates. 35 have been eaten. So in total they still have 74 - 35 = 39 chocolates. The answer is **39**.

Question: : Shawn has five toys. For Christmas, he got two toys each from his mom and dad. How many toys does he have now?
Step-by-Step Answer: He has 5 toys. He got 2 from mom, so after that he has 5 + 2 = 7 toys. Then he got 2 more from dad, so in total he has 7 + 2 = 9 toys. The answer is **9**.

Question: Michael had 58 golf balls. On Tuesday, he lost 23 golf balls. On Wednesday, he lost 2 more. How many golf balls did he have at the end of Wednesday?
Step-by-Step Answer: Michael started with 58 golf balls and lost 23, so he has 58 - 23 = 35. After he lost 2 more, he has 35 - 2 = 33 balls now. The answer is **33**.

Question: Olivia has $23. She bought five bagels for $3 each. How much money does she have left?
Step-by-Step Answer: 5 bagels for $3 each should cost 5 * 3 = 15 dollars. Olivia had $23 in the beginning, so now she has 23 - 15 = 8 dollars left. The answer is **8**.

Frage: *Janets Enten legen 16 Eier pro Tag. Sie isst drei jeden Morgen zum Frühstück und backt mit vier jeden Tag Muffins für ihre Freunde. Den Rest verkauft sie täglich auf dem Bauernmarkt für 2 US-Dollar pro frischem Entenei. Wie viel Dollar nimmt sie täglich auf dem Bauernmarkt ein?*
Step-by-Step Answer: *[...] The answer is* ***18***.

Multilingual Exemplars

Question: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
Step-by-Step Answer: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is **11**.

Frage: Es waren neun Computer im Serverraum. Von Montag bis Donnerstag wurden jeden Tag noch fünf Computer installiert. Wie viele Computer sind jetzt im Serverraum?
Step-by-Step Answer: There are 4 days from Monday to Thursday. 5 computers were added each day. That means in total 4 * 5 = 20 computers were added. There were 9 computers in the beginning, so now there are 9 + 20 = 29 computers. The answer is **29**.

Question: Léa avait 32 chocolats et sa sœur en avait 42. Si elles en ont mangé 35, combien de morceaux leur reste-t-il en tout ?
Step-by-Step Answer: Leah had 32 chocolates and Leah's sister had 42. That means there were originally 32 + 42 = 74 chocolates. 35 have been eaten. So in total they still have 74 - 35 = 39 chocolates. The answer is **39**.

Pregunta: Shawn tiene cinco juguetes. Para Navidad, recibió dos juguetes de su mamá y dos de su papá. ¿Cuántos juguetes tiene ahora?
Step-by-Step Answer: He has 5 toys. He got 2 from mom, so after that he has 5 + 2 = 7 toys. Then he got 2 more from dad, so in total he has 7 + 2 = 9 toys. The answer is **9**.

Задача: у Майкла было 58 мячей для гольфа. Во вторник он потерял 23 мяча для гольфа. В среду он потерял еще 2. Сколько мячей для гольфа осталось у него к концу среды?
Step-by-Step Answer: Michael started with 58 golf balls and lost 23, so he has 58 - 23 = 35. After he lost 2 more, he has 35 - 2 = 33 balls now. The answer is **33**.

问题： 奥利维亚有 23 美元。她买了五个单价 3 美元的百吉饼。 她还剩多少钱?
Step-by-Step Answer: 5 bagels for $3 each should cost 5 * 3 = 15 dollars. Olivia had $23 in the beginning, so now she has 23 - 15 = 8 dollars left. The answer is **8**.

Frage: *Janets Enten legen 16 Eier pro Tag. Sie isst drei jeden Morgen zum Frühstück und backt mit vier jeden Tag Muffins für ihre Freunde. Den Rest verkauft sie täglich auf dem Bauernmarkt für 2 US-Dollar pro frischem Entenei. Wie viel Dollar nimmt sie täglich auf dem Bauernmarkt ein?*
Step-by-Step Answer: *[...] The answer is* ***18***.

Figure 7: Prompt template in the English CoT setting (EN-CoT), solving a problem in German. Above dotted lines: few-shot exemplars; below dotted lines: the question of interest and the expected answer. The dotted lines are not included in our experiments.